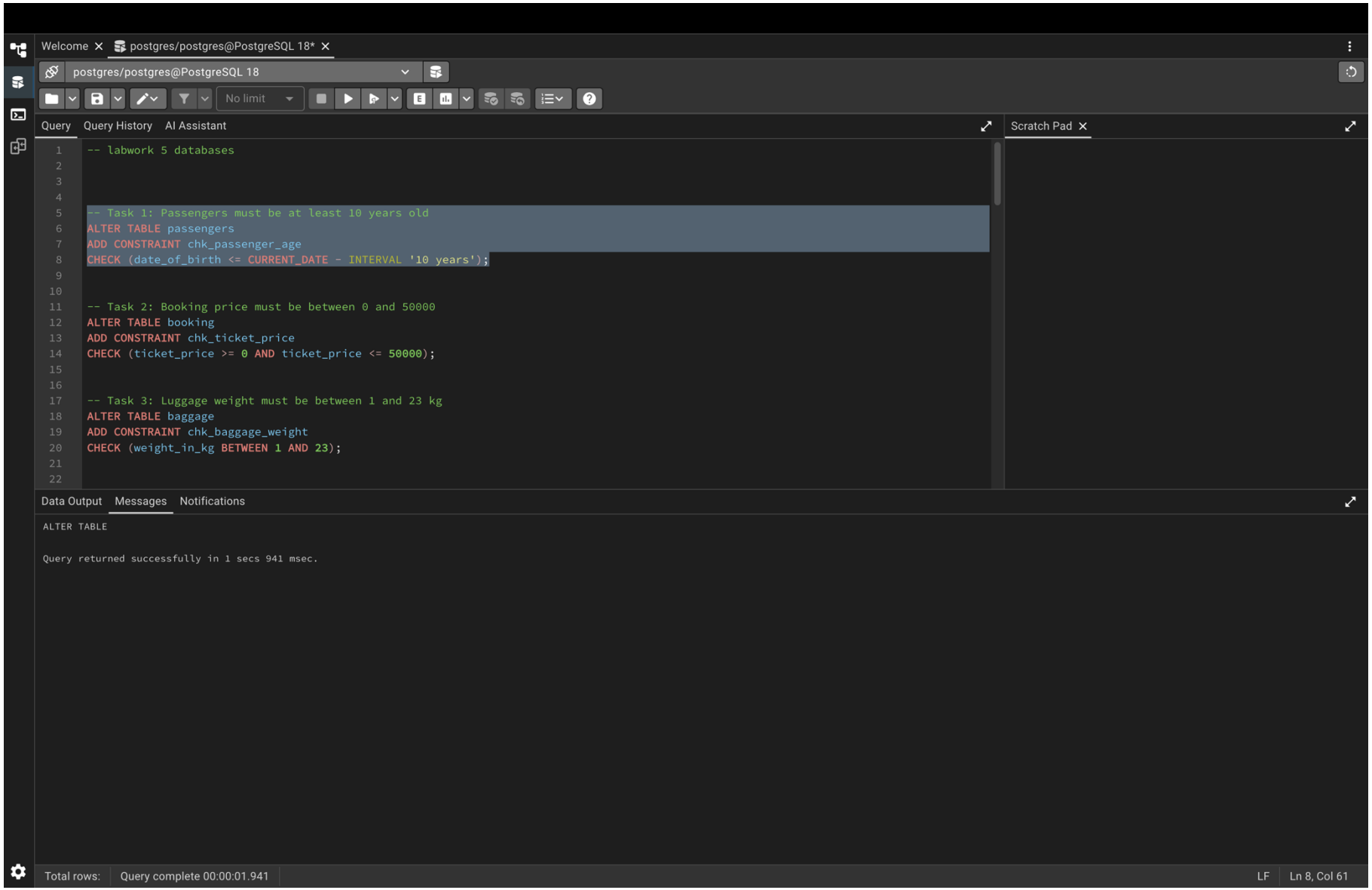
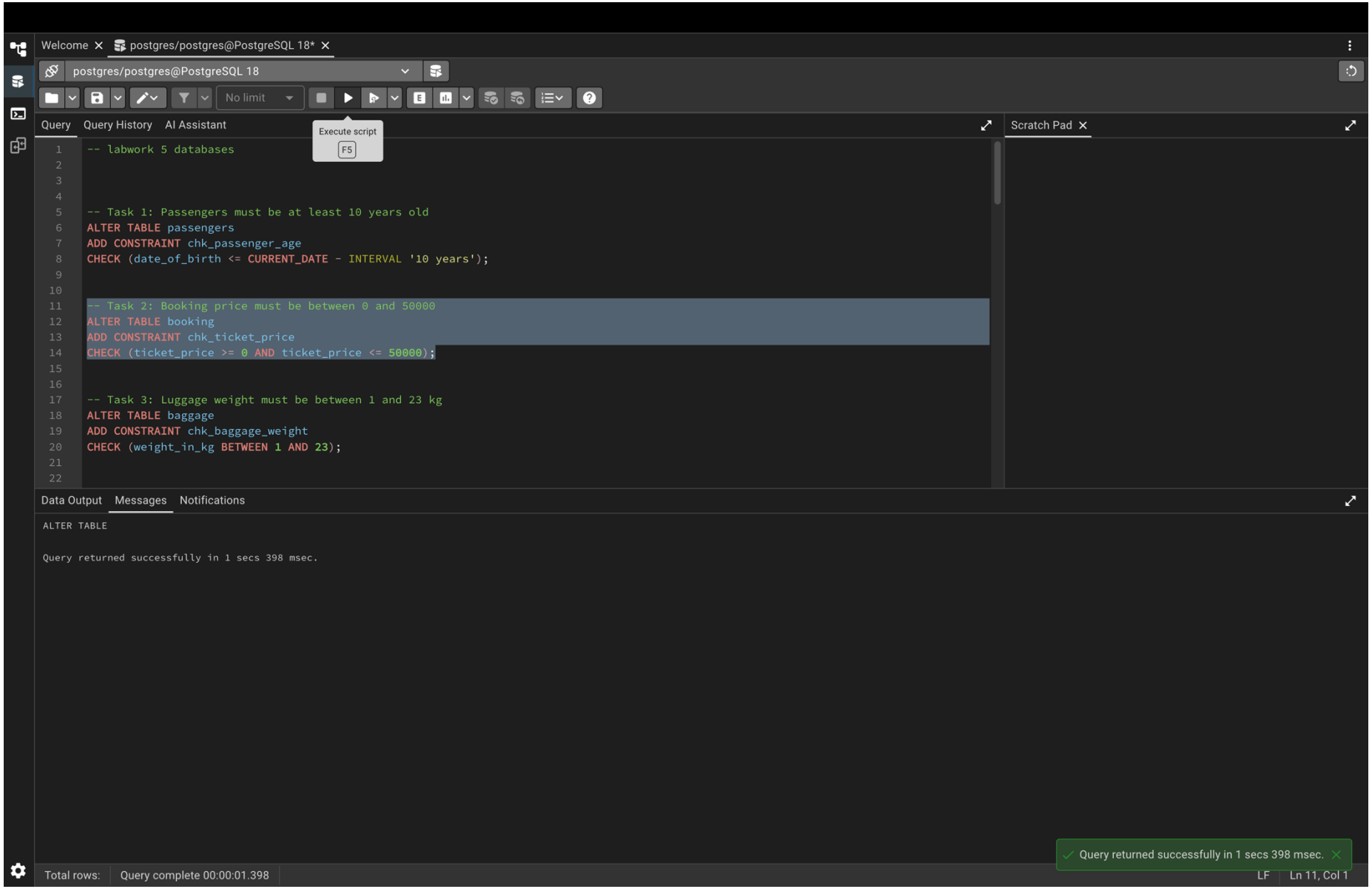
Labwork5 databases

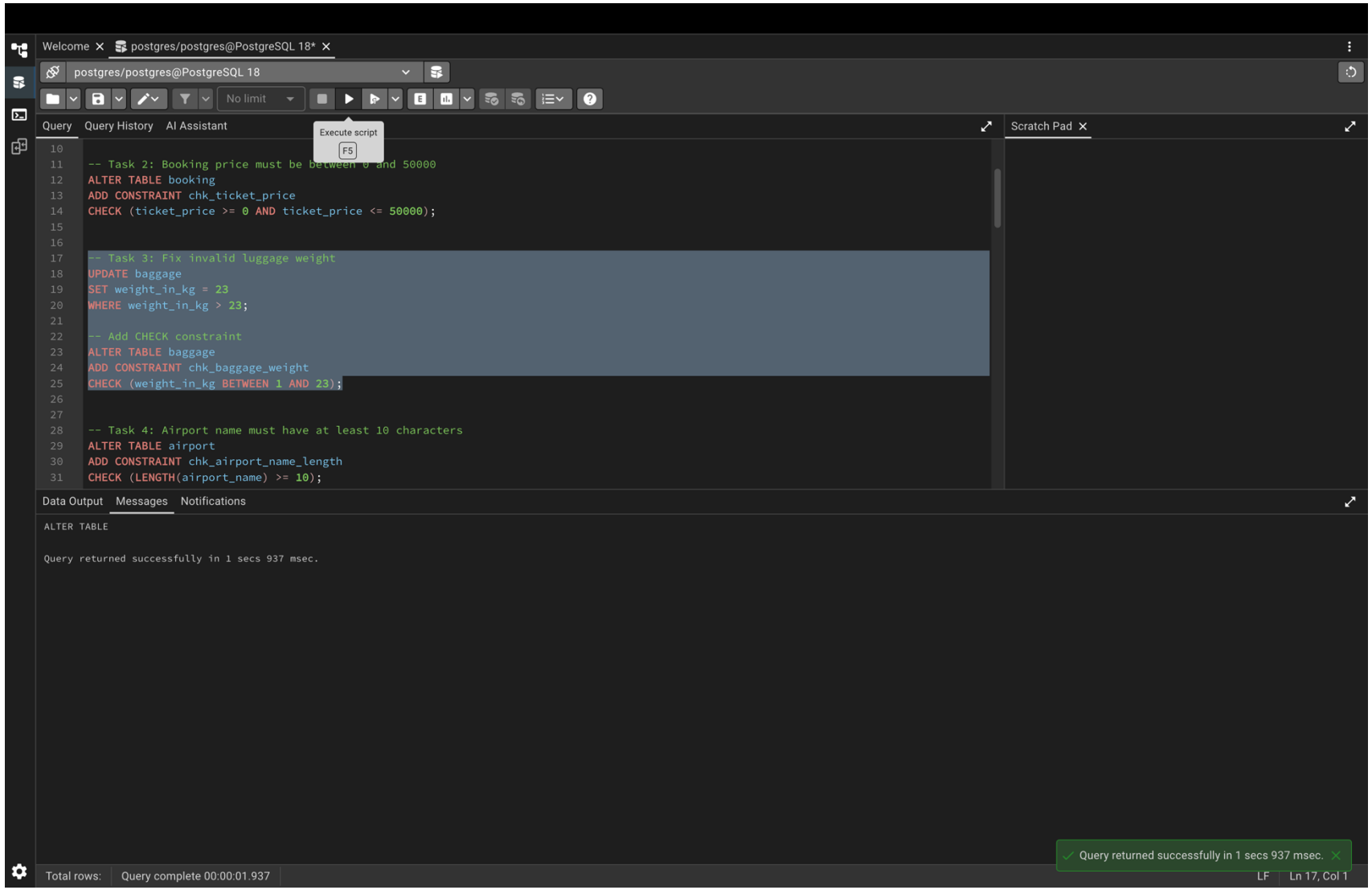
Bakytzhan Bekassyl


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
-- labwork 5 databases
-- Task 1: Passengers must be at least 10 years old
ALTER TABLE passengers
ADD CONSTRAINT chk_passenger_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
-- Task 2: Booking price must be between 0 and 50000
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_price
CHECK (ticket_price >= 0 AND ticket_price <= 50000);
-- Task 3: Luggage weight must be between 1 and 23 kg
ALTER TABLE baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 1 secs 941 msec.
Total rows:
Query complete 00:00:01.941
LF
Ln 8, Col 61

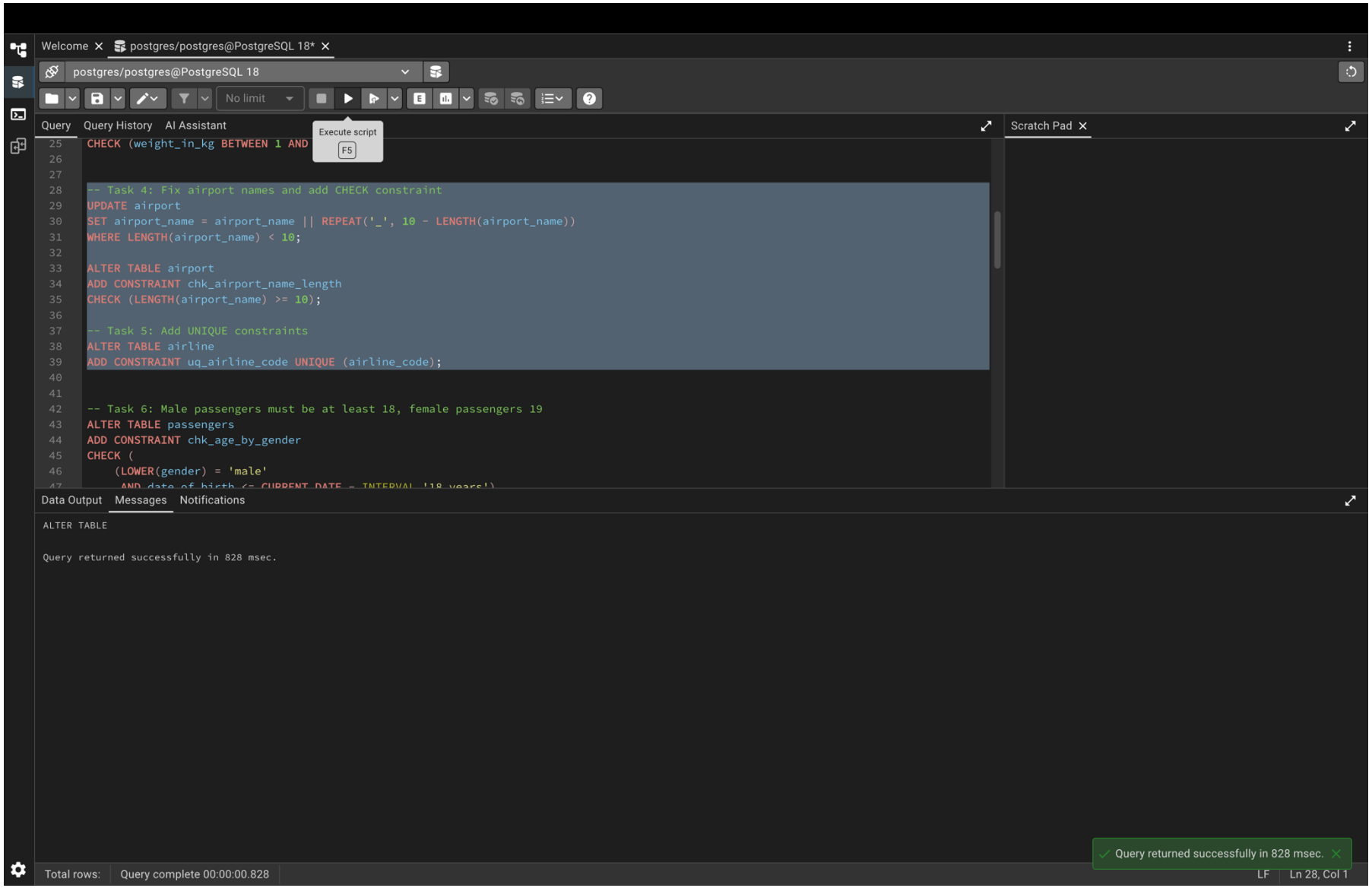
2


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
AI Assistant
Scratch Pad
-- labwork 5 databases
-- Task 1: Passengers must be at least 10 years old
ALTER TABLE passengers
ADD CONSTRAINT chk_passenger_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
-- Task 2: Booking price must be between 0 and 50000
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_price
CHECK (ticket_price >= 0 AND ticket_price <= 50000);
-- Task 3: Luggage weight must be between 1 and 23 kg
ALTER TABLE baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 1 secs 398 msec.
Query returned successfully in 1 secs 398 msec.
Total rows:
Query complete 00:00:01.398
LF
Ln 11, Col 1

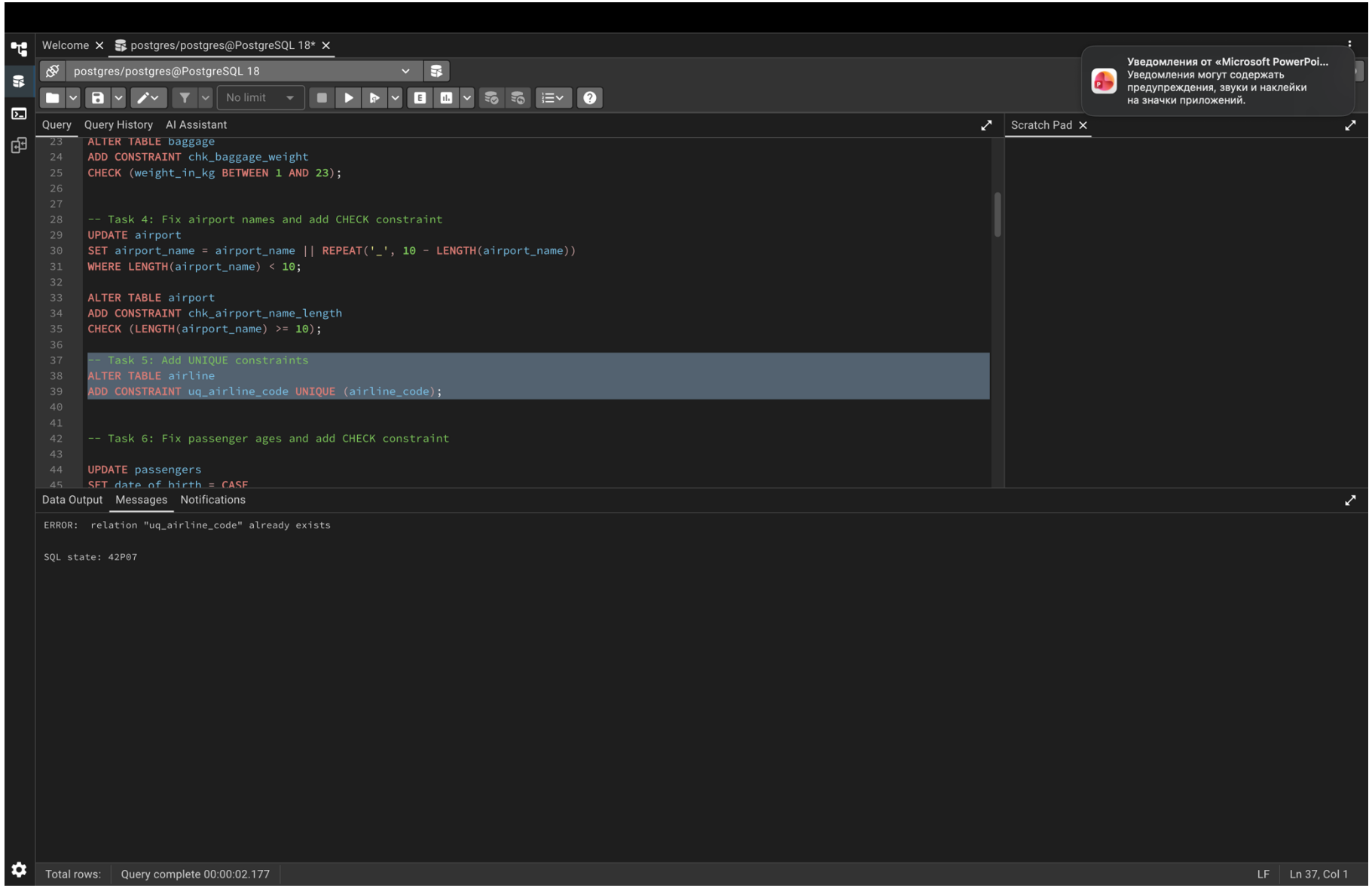
3


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
AI Assistant
Scratch Pad
10
11 -- Task 2: Booking price must be between 0 and 50000
12 ALTER TABLE booking
13 ADD CONSTRAINT chk_ticket_price
14 CHECK (ticket_price >= 0 AND ticket_price <= 50000);
15
16
17 -- Task 3: Fix invalid luggage weight
18 UPDATE baggage
19 SET weight_in_kg = 23
20 WHERE weight_in_kg > 23;
21
22 -- Add CHECK constraint
23 ALTER TABLE baggage
24 ADD CONSTRAINT chk_baggage_weight
25 CHECK (weight_in_kg BETWEEN 1 AND 23);
26
27
28 -- Task 4: Airport name must have at least 10 characters
29 ALTER TABLE airport
30 ADD CONSTRAINT chk_airport_name_length
31 CHECK (LENGTH(airport_name) >= 10);
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 1 secs 937 msec.
Query returned successfully in 1 secs 937 msec.
Total rows:
Query complete 00:00:01.937
LF
Ln 17, Col 1

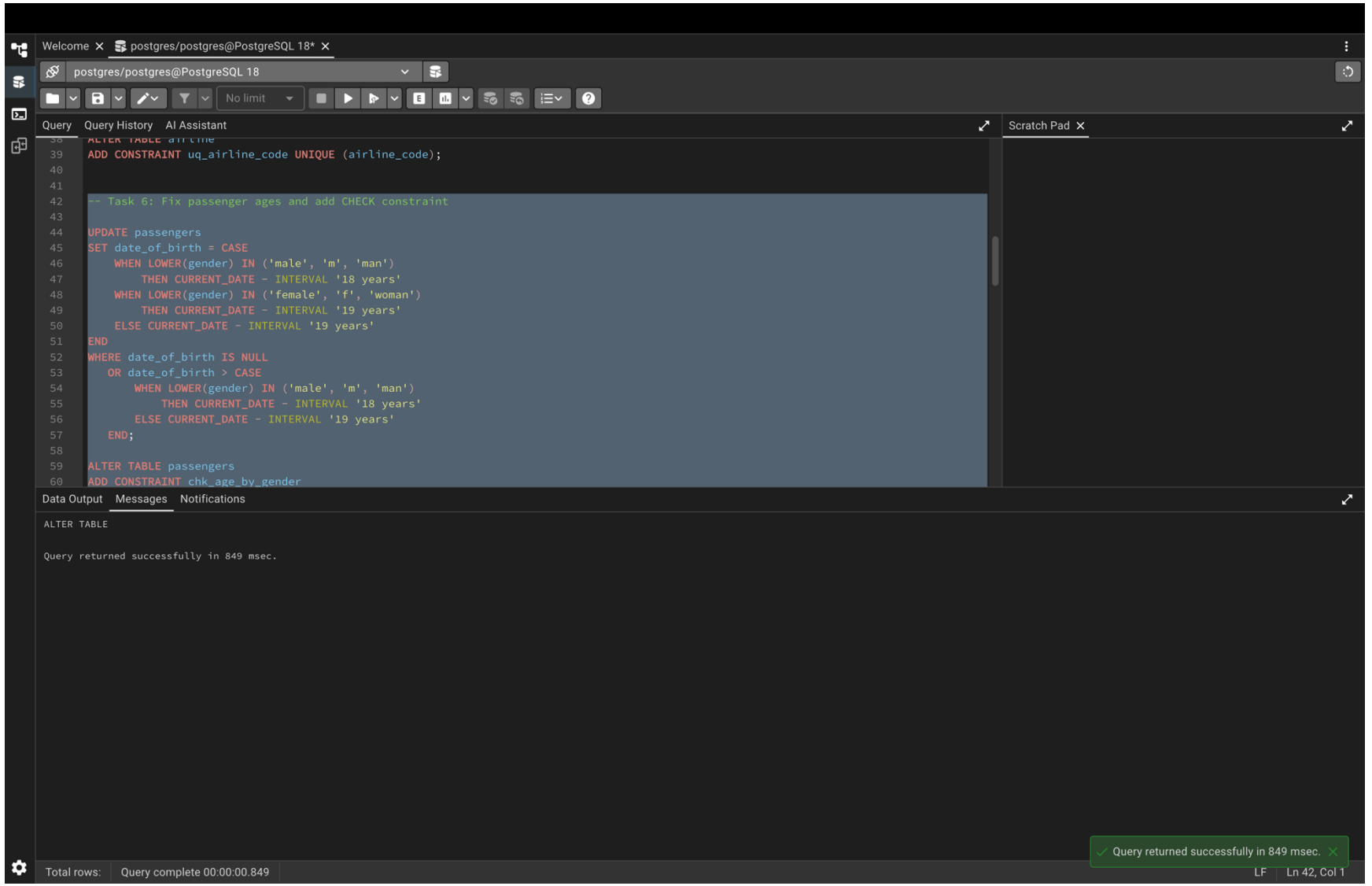
4


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Execute script
F5
Scratch Pad
CHECK (weight_in_kg BETWEEN 1 AND
-- Task 4: Fix airport names and add CHECK constraint
UPDATE airport
SET airport_name = airport_name || REPEAT('_', 10 - LENGTH(airport_name))
WHERE LENGTH(airport_name) < 10;
ALTER TABLE airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
-- Task 5: Add UNIQUE constraints
ALTER TABLE airline
ADD CONSTRAINT uq_airline_code UNIQUE (airline_code);
-- Task 6: Male passengers must be at least 18, female passengers 19
ALTER TABLE passengers
ADD CONSTRAINT chk_age_by_gender
CHECK (
(LOWER(gender) = 'male'
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 828 msec.
Query returned successfully in 828 msec.
Total rows:
Query complete 00:00:00.828
LF
Ln 28, Col 1

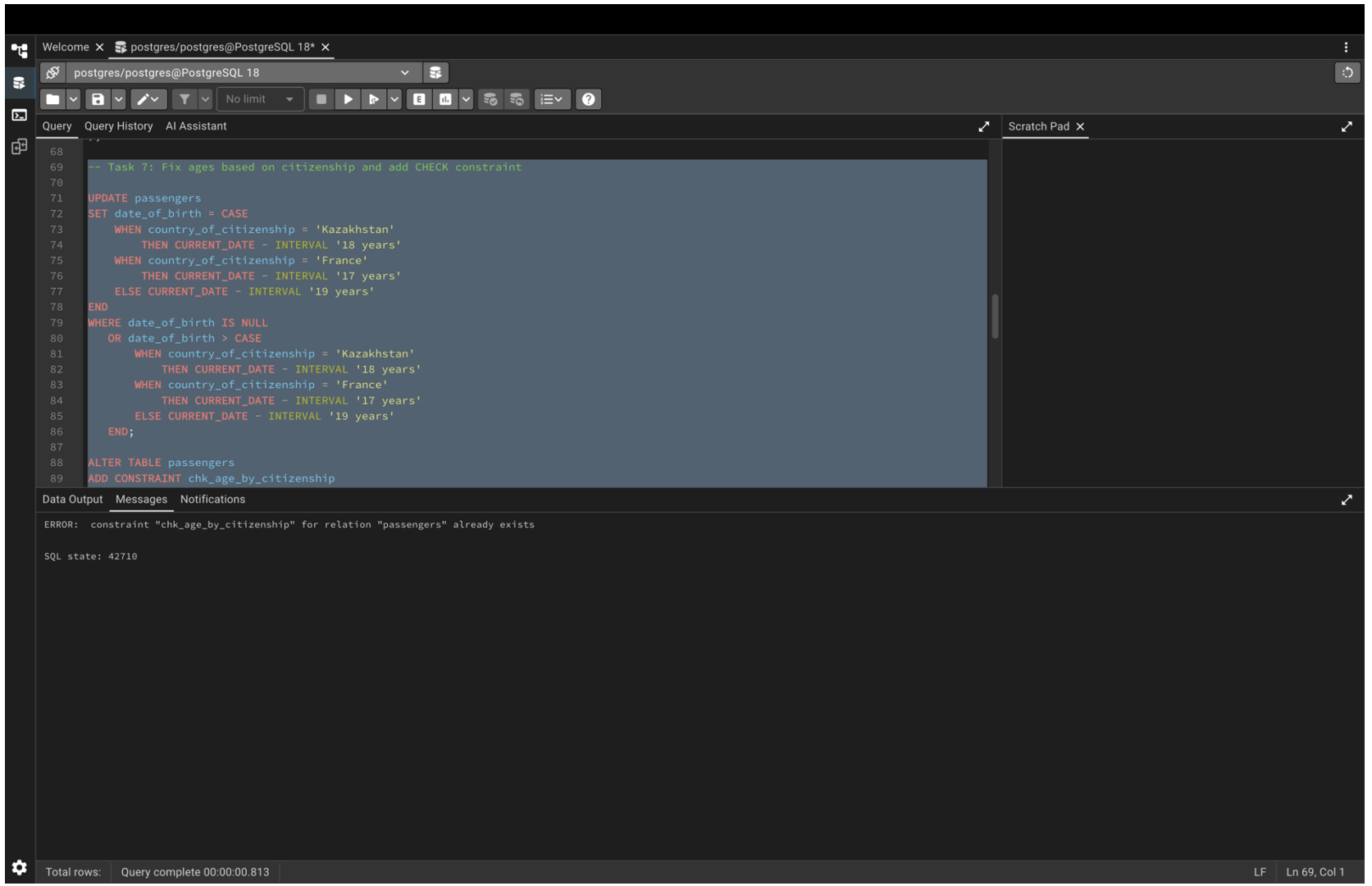
5


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Уведомления от «Microsoft PowerPoi...
Уведомления могут содержать предупреждения, звуки и наклейки на значки приложений.
Query
Query History
AI Assistant
Scratch Pad
ALTER TABLE baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
-- Task 4: Fix airport names and add CHECK constraint
UPDATE airport
SET airport_name = airport_name || REPEAT('_', 10 - LENGTH(airport_name))
WHERE LENGTH(airport_name) < 10;
ALTER TABLE airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
-- Task 5: Add UNIQUE constraints
ALTER TABLE airline
ADD CONSTRAINT uq_airline_code UNIQUE (airline_code);
-- Task 6: Fix passenger ages and add CHECK constraint
UPDATE passengers
SET date_of_birth = CASE
Data Output
Messages
Notifications
ERROR: relation "uq_airline_code" already exists
SQL state: 42P07
Total rows:
Query complete 00:00:02.177
LF
Ln 37, Col 1

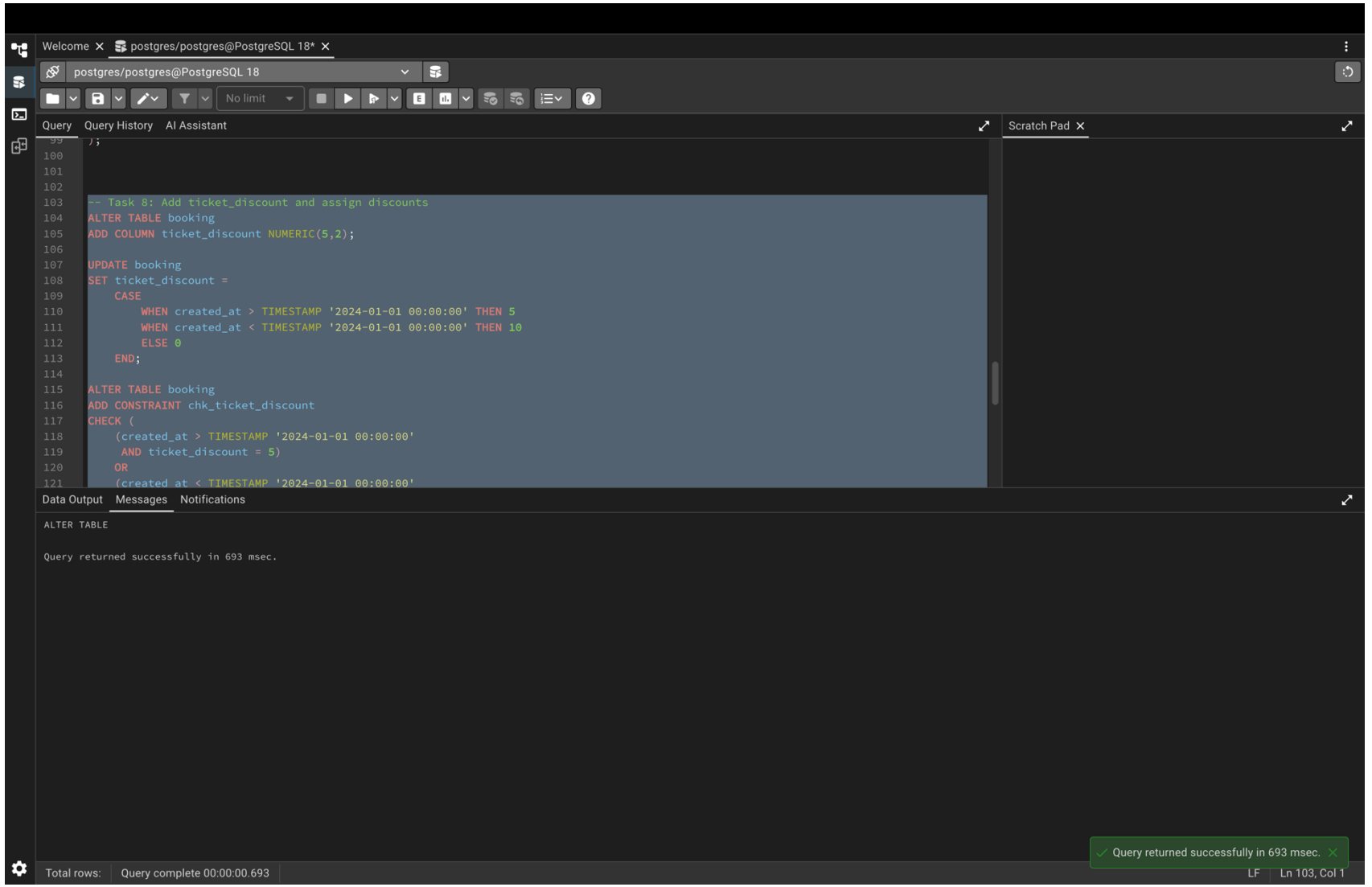
6


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
ALTER TABLE airline
ADD CONSTRAINT uq_airline_code UNIQUE (airline_code);
-- Task 6: Fix passenger ages and add CHECK constraint
UPDATE passengers
SET date_of_birth = CASE
WHEN LOWER(gender) IN ('male', 'm', 'man')
THEN CURRENT_DATE - INTERVAL '18 years'
WHEN LOWER(gender) IN ('female', 'f', 'woman')
THEN CURRENT_DATE - INTERVAL '19 years'
ELSE CURRENT_DATE - INTERVAL '19 years'
END
WHERE date_of_birth IS NULL
OR date_of_birth > CASE
WHEN LOWER(gender) IN ('male', 'm', 'man')
THEN CURRENT_DATE - INTERVAL '18 years'
ELSE CURRENT_DATE - INTERVAL '19 years'
END;
ALTER TABLE passengers
ADD CONSTRAINT chk_age_by_gender
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 849 msec.
Query returned successfully in 849 msec.
Total rows:
Query complete 00:00:00.849
LF
Ln 42, Col 1

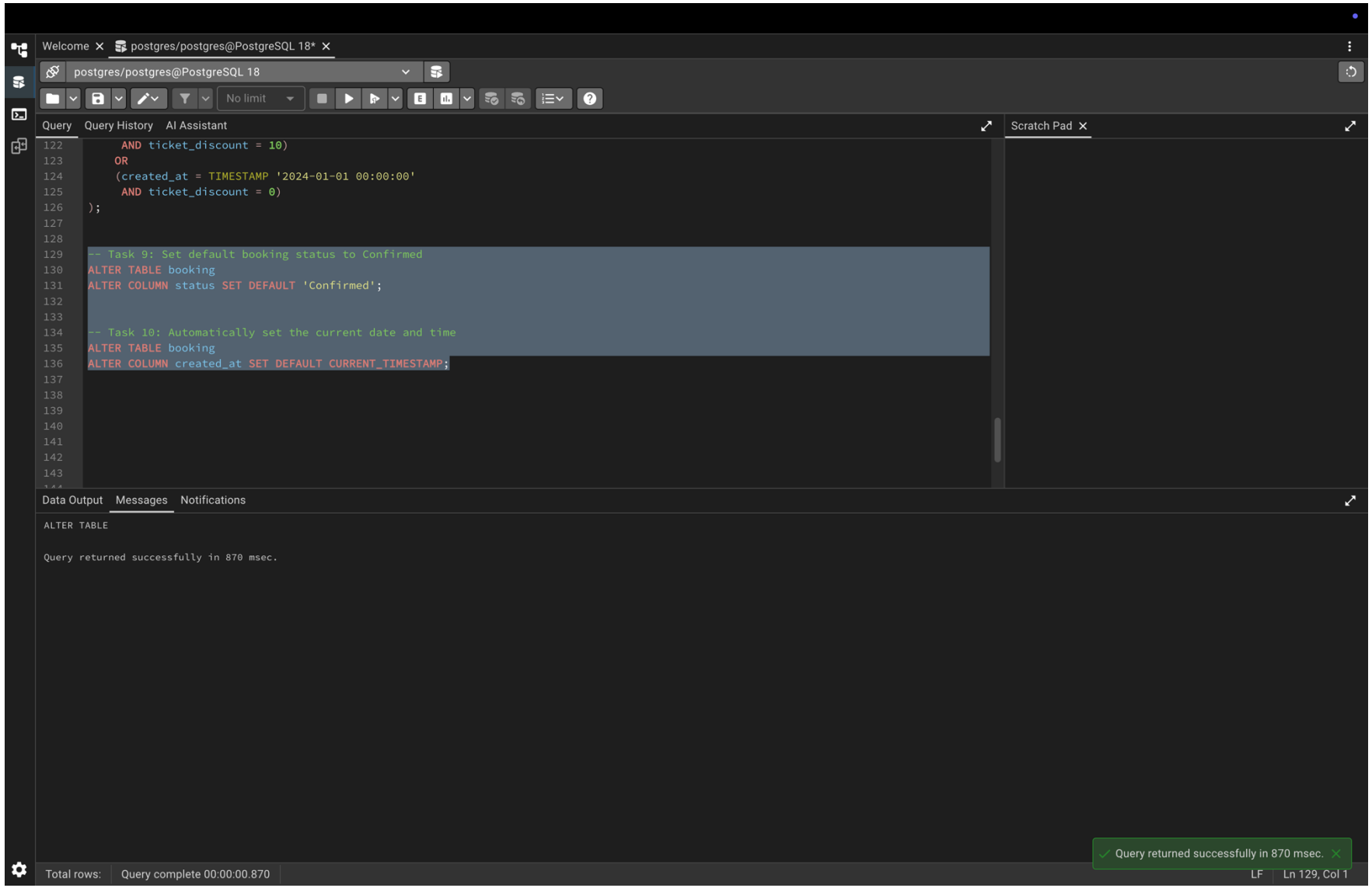
7


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
-- Task 7: Fix ages based on citizenship and add CHECK constraint
UPDATE passengers
SET date_of_birth = CASE
WHEN country_of_citizenship = 'Kazakhstan'
THEN CURRENT_DATE - INTERVAL '18 years'
WHEN country_of_citizenship = 'France'
THEN CURRENT_DATE - INTERVAL '17 years'
ELSE CURRENT_DATE - INTERVAL '19 years'
END
WHERE date_of_birth IS NULL
OR date_of_birth > CASE
WHEN country_of_citizenship = 'Kazakhstan'
THEN CURRENT_DATE - INTERVAL '18 years'
WHEN country_of_citizenship = 'France'
THEN CURRENT_DATE - INTERVAL '17 years'
ELSE CURRENT_DATE - INTERVAL '19 years'
END;
ALTER TABLE passengers
ADD CONSTRAINT chk_age_by_citizenship
Data Output
Messages
Notifications
ERROR: constraint "chk_age_by_citizenship" for relation "passengers" already exists
SQL state: 42710
Total rows:
Query complete 00:00:00.813
LF
Ln 69, Col 1

8


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
-- Task 8: Add ticket_discount and assign discounts
ALTER TABLE booking
ADD COLUMN ticket_discount NUMERIC(5,2);
UPDATE booking
SET ticket_discount =
CASE
WHEN created_at > TIMESTAMP '2024-01-01 00:00:00' THEN 5
WHEN created_at < TIMESTAMP '2024-01-01 00:00:00' THEN 10
ELSE 0
END;
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_discount
CHECK (
(created_at > TIMESTAMP '2024-01-01 00:00:00'
AND ticket_discount = 5)
OR
(created_at < TIMESTAMP '2024-01-01 00:00:00'
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 693 msec.
Query returned successfully in 693 msec.
Total rows:
Query complete 00:00:00.693
LF
Ln 103, Col 1

9-10


Welcome
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
AND ticket_discount = 10)
OR
(created_at = TIMESTAMP '2024-01-01 00:00:00'
AND ticket_discount = 0)
);
-- Task 9: Set default booking status to Confirmed
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
-- Task 10: Automatically set the current date and time
ALTER TABLE booking
ALTER COLUMN created_at SET DEFAULT CURRENT_TIMESTAMP;
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 870 msec.
Query returned successfully in 870 msec.
Total rows:
Query complete 00:00:00.870
LF
Ln 129, Col 1